

```

import numpy as np
import matplotlib.pyplot as plt

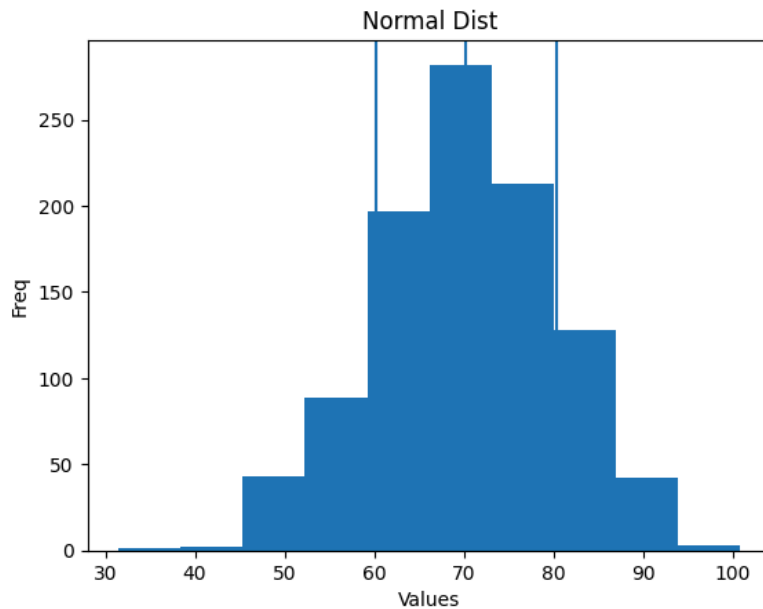
data=np.random.normal(70,10,1000)
plt.hist(data)
mean=np.mean(data)
std=np.std(data)

plt.axvline(mean)
plt.axvline(mean+std)
plt.axvline(mean-std)
plt.title("Normal Dist")
plt.xlabel("Values")
plt.ylabel("Freq")
plt.show()

count=0
for value in data:
    # check if the score lies in the given range
    if value >=60 and value <=80:
        count+=1

percentage=count/len(data)
print("percentage:", percentage)

```



percentage: 0.67

```

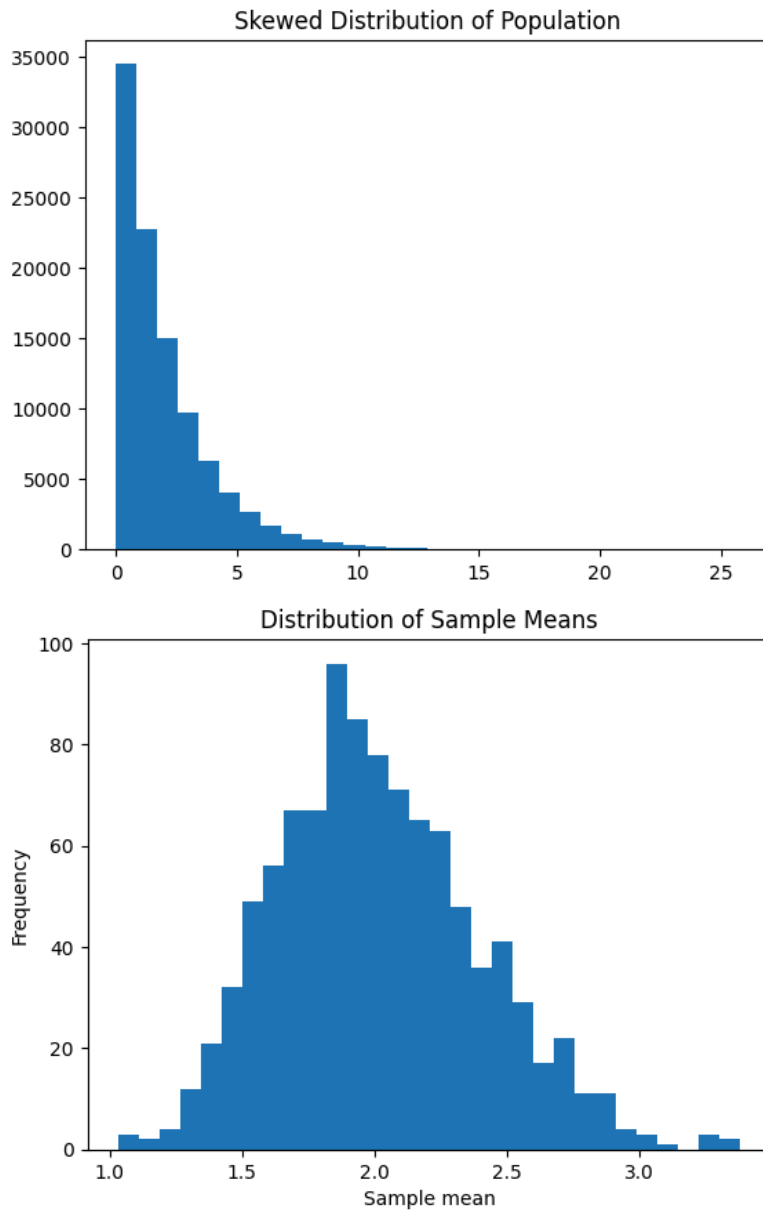
# CLT

population=np.random.exponential(scale=2, size=100000)
plt.title("Skewed Distribution of Population")
plt.hist(population, bins=30)
plt.show()
sample_means=[]

# take many samples
for i in range(1,1000):
    sample=np.random.choice(population, size=30)
    sample_mean=np.mean(sample)
    sample_means.append(sample_mean)

plt.hist(sample_means, bins=30)
plt.title("Distribution of Sample Means")
plt.xlabel("Sample mean")
plt.ylabel("Frequency")
plt.show()

```



```
import numpy as np
import matplotlib.pyplot as plt

# Skewed population
population = np.random.exponential(scale=2, size=100000)

# Different sample sizes
sample_sizes = [2, 5, 10, 30]

for n in sample_sizes:
    sample_means = []

    # take many samples
    for i in range(1000):
        sample = np.random.choice(population, size=n)
        sample_mean = np.mean(sample)
        sample_means.append(sample_mean)

    # plot histogram
    plt.hist(sample_means, bins=30, alpha=0.5, label="n = " + str(n))

plt.title("CLT: Sample Means for Different Sample Sizes")
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.legend()
plt.show()
```

